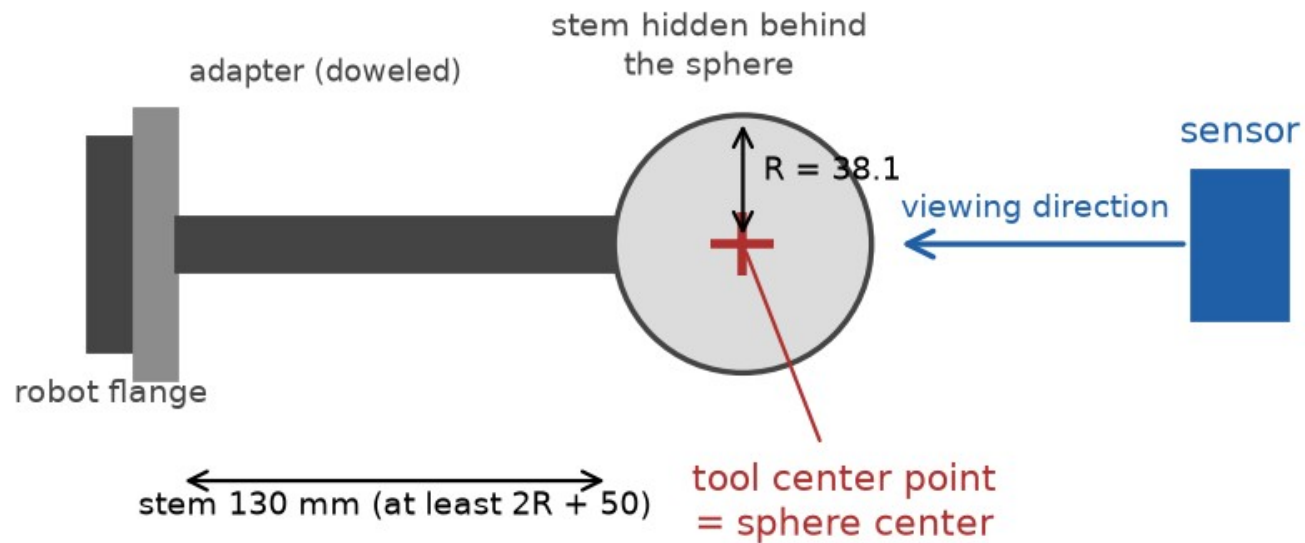


Stage-1 Calibration: Procurement and Build

The spheres, board and fixtures needed for the depth-sensor calibration captures



Two spheres and a board carry the known geometry



Sphere on its stem, side view: the stem points away from the sensor so that the sphere hides it.



Sphere A

76.2 mm diameter; used from 300 to 550 mm



Sphere B

152.4 mm diameter; used from 500 to 1,100 mm



Board

200 x 150 mm, at least 6 mm thick, flat to 0.05 mm after painting; tilted 0, 20 and 40 degrees

What must be built

Item	Quantity	Drawing
Sphere A with mounting interface	1	SC1-03 (hole detail)
Sphere B with mounting interface	1	SC1-04 (interface detail)
Sphere-flange mounts: adapter plates and stems	2 + 2	SC1-01 to SC1-04
Flat target (board)	1	SC1-05 (outline)
Flat-target mount (board adapter)	1	SC1-05
Three-ball bed (nest)	1	SC1-06
Run-out fixture: dial indicator on a magnetic base and steel base plate	1	none needed
Rigid sensor mount	1	existing drawing



Section 1b of the procedure; nothing outside sections 1a to 1f is needed for a session.

What must be bought

Item	Purpose	Estimated cost (USD)
Outside micrometers 75-100 and 150-175 mm, with setting standards	Sphere diameters, measured in-house	\$405 to \$520
Calipers with depth rod, 150 mm	Rough TCP length; board distance D	\$30 to \$150
Straightedge 300 mm and feeler gauges	Board flatness; stem straightness	\$75 to \$240
Thermometer	Room temperature at measurement	\$15 to \$40
Machinist's square	Board x axis	\$30 to \$80
Torque wrench	Recorded stem and adapter torques	\$80 to \$250
Consumables	Thread locker, retaining compound, shims, alcohol, padded cases	\$50 to \$150

A phone camera is enough for the setup photos. Already in hand: sensor, robot, capture computer with its communication software, analysis computer.

Building and buying cost about \$2,525 to \$6,500

Item to build	Estimated cost (USD)
Sphere A with mounting interface	\$250 to \$550
Sphere B with mounting interface	\$500 to \$1,500
Sphere-flange mounts: two adapter plates and two stems	\$400 to \$1,000
Flat target (board)	\$50 to \$400
Flat-target mount (board adapter)	\$250 to \$600
Three-ball bed (nest)	\$150 to \$450
Run-out fixture: dial indicator, magnetic base, steel base plate	\$240 to \$570
Rigid sensor mount	not estimated

\$2,525 to \$6,500

bearing-ball sphere A, aluminum sphere B,
tools included

\$4,375 to \$11,950

both spheres made to order, tools included

Tools and consumables \$685 to \$1,430, included in the totals. Estimates of October 2026, plus or minus 50 percent; sphere B is the least certain.

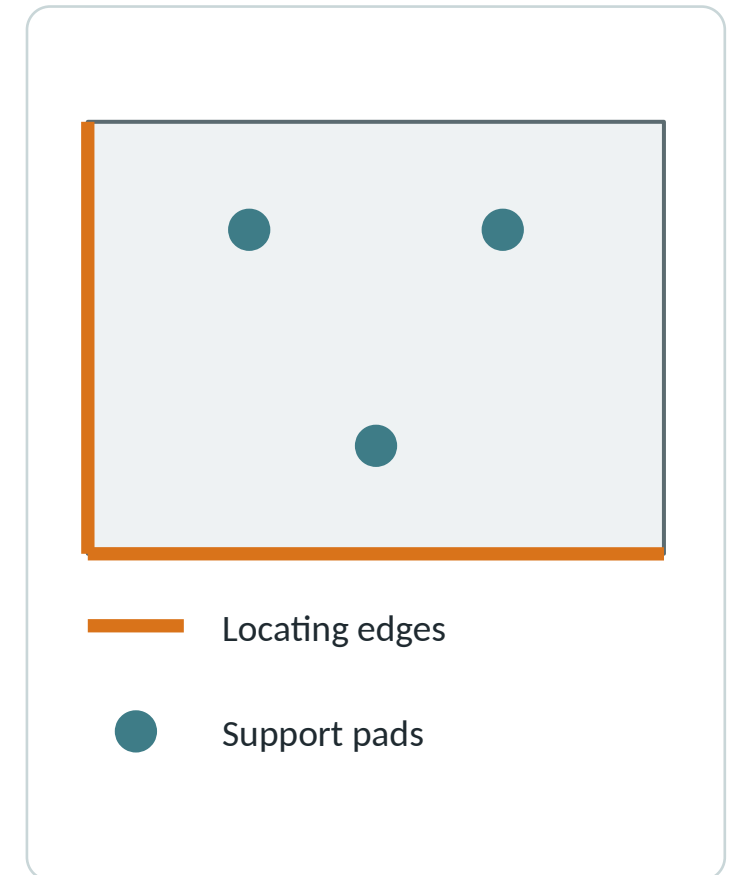
Purchase specification: target spheres

Requirement	Sphere A	Sphere B
Nominal diameter	76.2 mm (3 inch)	152.4 mm (6 inch)
Diameter	Known to 0.025 mm, by micrometer at six or more orientations after finishing	Same; measure at the session's room temperature
Sphericity	0.02 mm or better	0.02 mm or better
Material	Bearing-grade steel (52100 or 440C, grade 25 or better) or ceramic	Turned aluminum (6061-T6), solid or hollow with 6 mm wall
Surface	Matte, light, uniform: fine glass-bead blast or thin matte paint	Matte hard anodize, same appearance as sphere A
Mounting interface	Steel: blind hole 10 H7 x 23 mm, stem bonded. Ceramic: maker's M8 insert	M20 x 30 mm tapped hole at a pole, 32 mm spot face
Approximate mass	0.9 to 1.8 kg	about 5 kg

Procedure section 1d; drawings SC1-03 and SC1-04 show the interfaces.

Purchase specification: flat plate

Requirement	Flat plate (board)
Size	200 x 150 mm, at least 6 mm thick
Material	Finish-ground aluminum tooling plate (for example MIC-6) or float glass
Flatness	0.05 mm over the front face, checked with a straightedge and a 0.05 mm feeler leaf after painting and after mounting
Front face	Thin, even matte light-gray paint; do not bead-blast a plate thinner than about 10 mm
Edges	Bottom long edge and left short edge square and clean: they rest on the adapter's edge pins
Documents	Supplier's flatness report, kept with the session notes



Procedure section 1d; the board outline is on drawing SC1-05.

Spheres must be matte and measured, not certified



Finish

- Every bearing ball arrives mirror-bright: unusable
- Bead-blast with fine glass bead: removes micrometers
- Matte paint is second choice: 0.02 to 0.05 mm per coat
- "Chrome steel" is the 52100 alloy, not plating



Diameter

- Known to 0.025 mm, measured after the finish
- Measured in-house by outside micrometer at six or more orientations
- Aluminum grows 0.0035 mm per degree C: measure at session temperature
- The mean goes in the pose log; the spread checks sphericity



Mounting

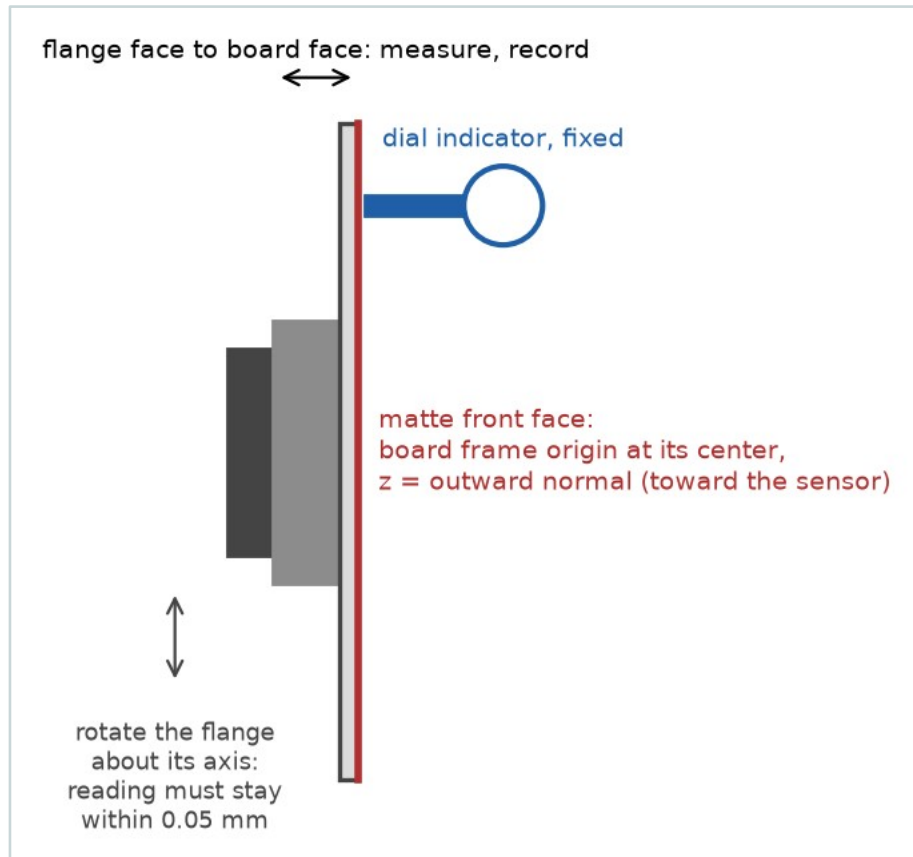
- Hardened balls cannot be tapped
- Steel ball: drilled or EDM blind hole, bonded stem
- Aluminum sphere B: thread cut when it is turned
- No pre-tapped precision sphere exists above 1.5 inch

Candidate suppliers

Item	Candidate suppliers	Notes
Spheres, made to order	Bal-tec; RGP Balls; Industrial Tectonics; Morgan Advanced Materials	3 and 6 inch matte spheres with a mounting hole are custom items
Matte reference spheres, stock	MetrologyWorks; Rothbucher RSLB10M (Baseline Equipment)	MetrologyWorks up to 1.5 inch; Rothbucher 145 mm, sphericity 0.3 mm
Board plate	McMaster-Carr MIC-6 plus a grinding shop; Edmund Optics float glass	Painted matte after grinding; flatness checked after mounting
Nest balls	McMaster-Carr; Bal-tec	Grade-25 bearing balls
Adapters, stems, sphere holes	Local machine shop; EDM shop for hardened balls	Hardened balls cannot be tapped
Diameter measurement	Outside micrometers: Mitutoyo 103-140-10 and 103-143-10, or Starrett	In-house, after the matte finish; setting standards included
Dial indicator and magnetic base	Mitutoyo 2046 series; Noga MG71003 or DG-61003	For the run-out fixture

From suppliers' web pages, October 2026; not an endorsement. Confirm size, finish, measurement and thread before ordering.

Run-out fixture: dial indicator and its mounting



The run-out check: the flange turns, the indicator stays still, the reading must stay within 0.05 mm.



Dial indicator

0.01 mm graduation, about 10 mm travel; for example Mitutoyo 2046 series, about \$50 to \$150



Magnetic base

Articulating, with fine adjustment; for example Noga MG71003 or DG-61003, about \$140 to \$320



Steel base plate

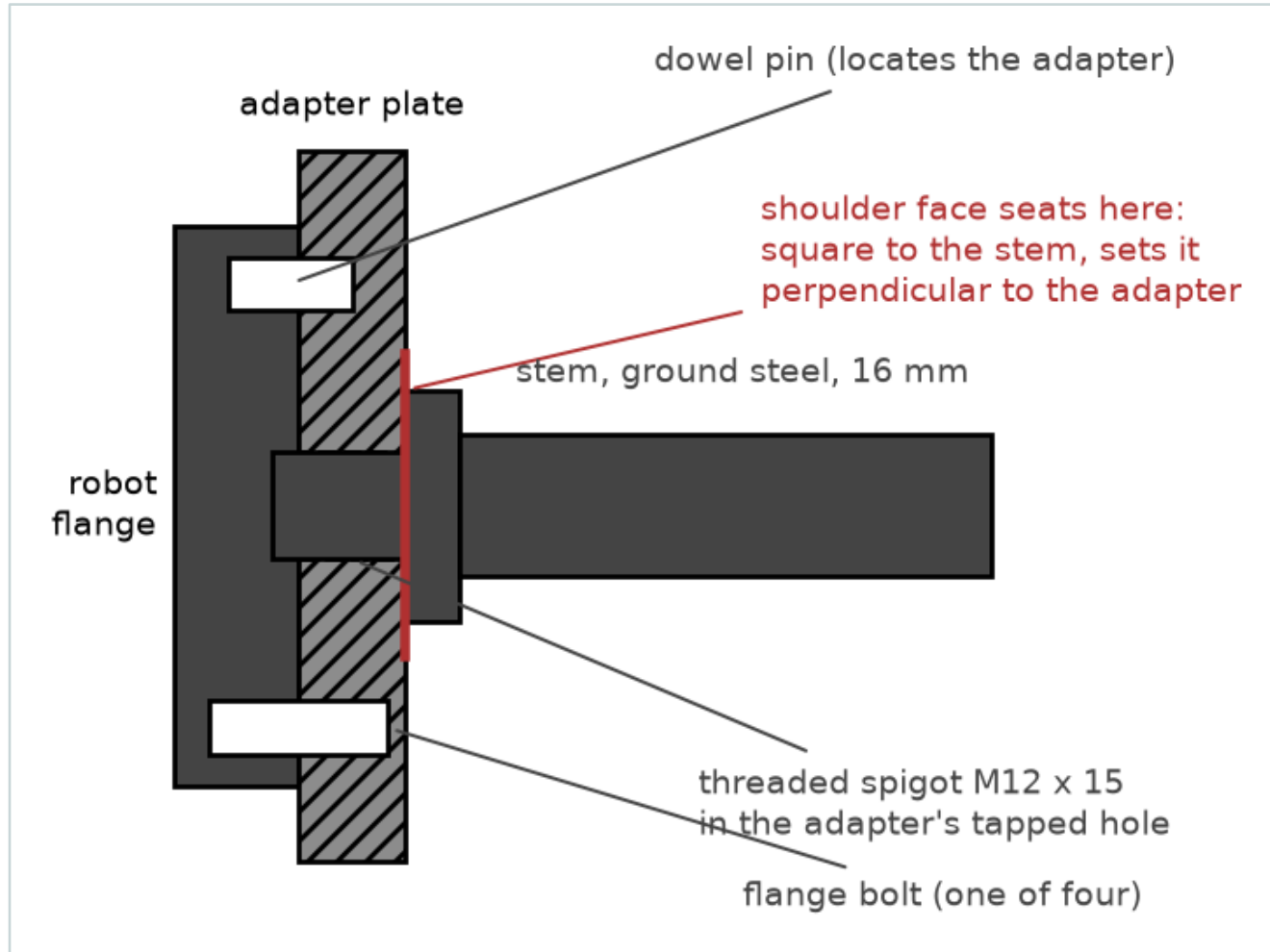
About 150 x 100 x 12 mm, bolted to the cell table with two M8 screws, so the base holds on any table and does not creep



Placement

Where the indicator tip reaches the board's front face about 20 mm from an edge with the robot in its run-out pose

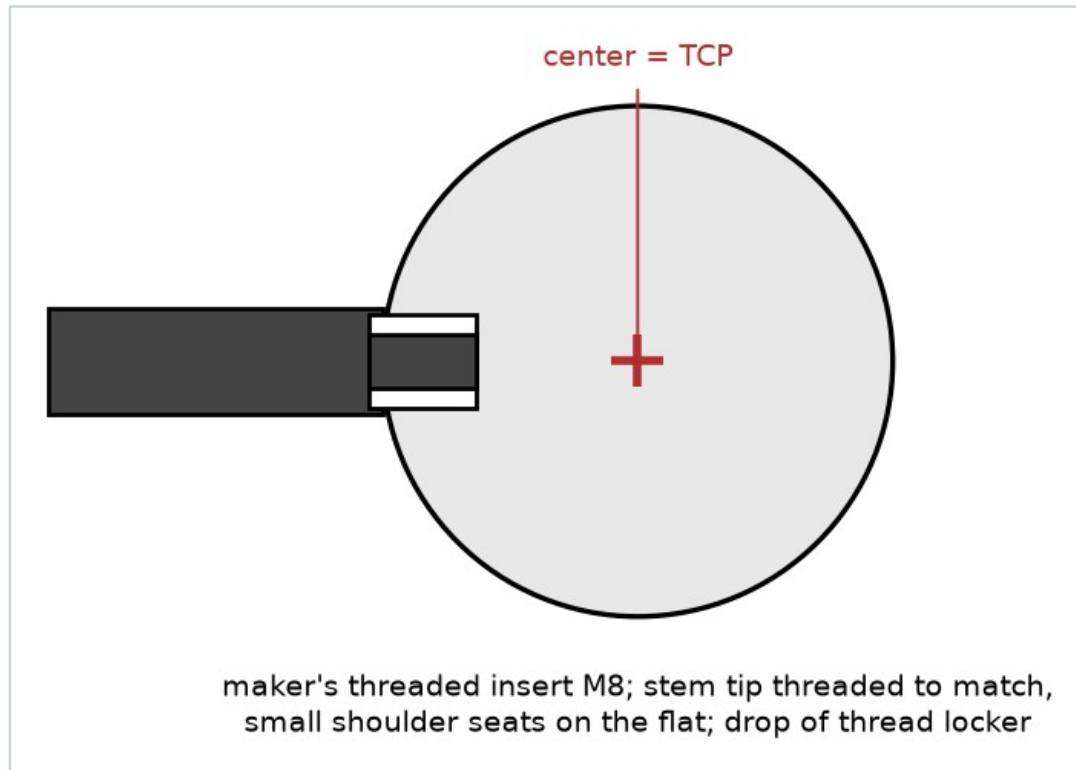
A stiff, repeatable chain from flange to sphere



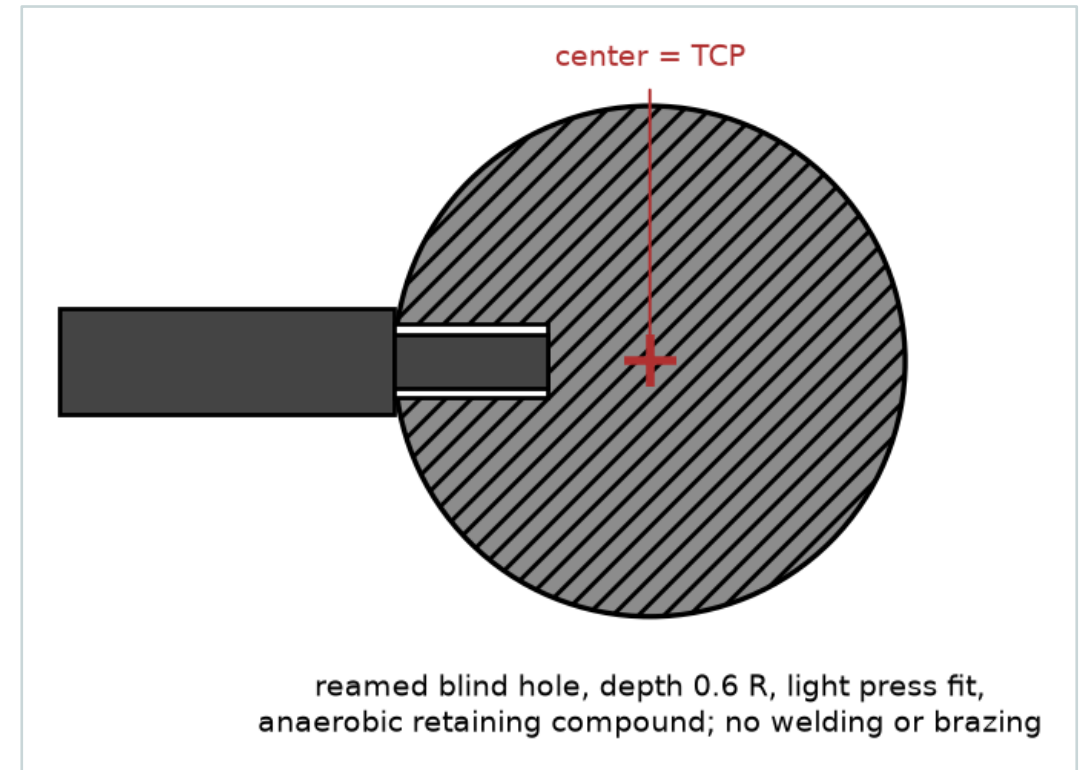
- Adapter located by the flange dowel, so a remount lands in the same place
- Ground stem, 16 mm for A and 30 mm for B, keeps sag to 0.02 to 0.05 mm
- Stem length at least $2R + 50$ mm, so the sphere hides the stem
- Witness mark across stem and adapter shows a loosened joint
- After any remount, re-check in the nest: readings within 0.1 mm

Adapter on the flange's dowel, with the stem's shoulder seated on the adapter face.

The sphere end: threaded insert or bonded stem



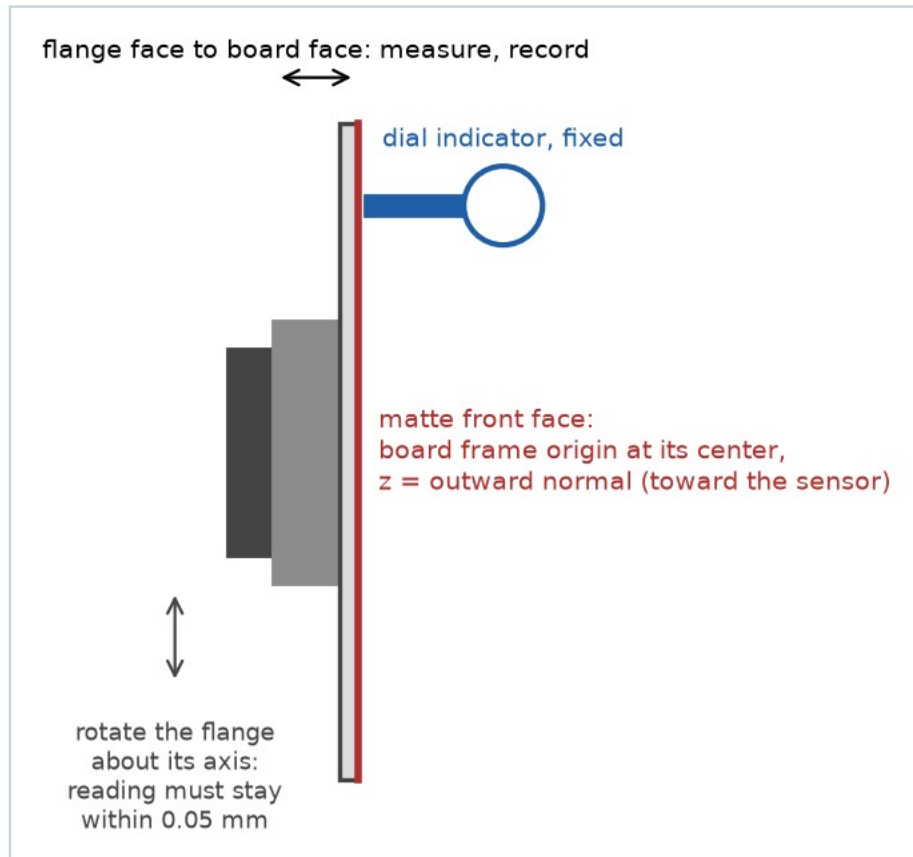
Ceramic sphere: maker's threaded insert



Steel sphere: bonded blind hole

The hole need not be exactly radial: the tool-center-point routine measures the center wherever it ends up.

The board must stay flat after finishing and mounting



Board on its adapter. The runout and frame checks shown here are done at setup (deck 2).



Plate

200 x 150 mm, at least 6 mm thick; ground aluminum tooling plate or float glass



Finish

Matte light gray paint on the front face; do not bead-blast a plate thinner than about 10 mm, it bows



Flatness

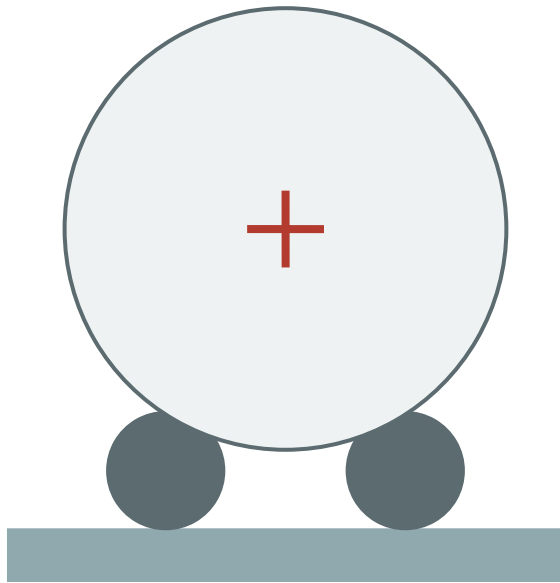
0.05 mm, checked after painting and mounting, not before; keep the flatness report



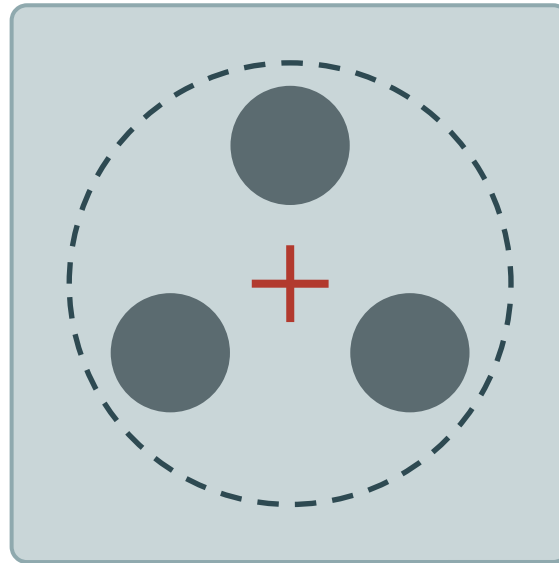
Adapter

Three support pads, three edge pins and three clamp fingers hold the board square to the flange axis, centered on it, and in the same place after a remount

A three-ball nest holds each sphere at a fixed center



Side view



Top view



Three hardened balls, 24 mm diameter, bonded into pockets on a 60 mm circle in a 120 mm steel base



Ordinary grade-25 bearing balls are good enough: the nest's quality comes from the balls being rigidly fixed



Bolted to the table where the robot reaches with the wrist down and tilted about 40 degrees to either side



Used once per sphere at setup, and after any remount of an adapter or stem

Checks before the fixtures go to the robot cell

- ✓ Adapter plates, stems, board adapter and nest base made to drawings SC1-01 to SC1-06
- ✓ Adapter thread square to its stem-side face within 0.02 mm over 100 mm
- ✓ Each stem straight: check against the straightedge
- ✓ Both spheres matte, uniform and free of highlights
- ✓ Each sphere diameter measured by micrometer after finishing, spread within 0.02 mm, temperature recorded
- ✓ Board flat to 0.05 mm after painting and mounting
- ✓ Each fixture weighed for its tool load data; witness marks across each stem-adapter joint

